SIEMENS-DTU ELECTIVE PROGRAM

System Verilog MCQ

Candidate Information

Field	Details
Student Name	_____
Roll Number	_____
Branch / Semester	_____
Date	_____

Instructions:

- Number of Questions: 15
- Time: 45 minutes
- 4 marks for correct answer and -1 for incorrect answer

Q1. Which of the following statements about logic and bit datatypes is correct?

A) logic is a 2-state type; bit is a 4-state type

B) logic is a 4-state type; bit is a 2-state type

C) Both logic and bit are 4-state types with identical behavior

D) bit cannot be used in testbenches, only in RTL modules

Q2. What are the values of s and u respectively?

int unsigned u = -5;

int signed s = -5;

A) s = -5, u = -5

B) s = -5, u = 4294967291

C) s = 4294967291, u = -5

D) Both are implementation-dependent

Q3. What is the key difference?

```
typedef struct {
  int a;
  int b;
} my_struct;
typedef union {
  int a;
  int b;
} my_union;
```

A) Both allocate memory for all members

B) struct allocates memory for all members; union shares memory among members

C) union allocates more memory than struct

D) They are functionally identical

Q4. What is printed?

```
byte b = -1;
int i = -1;
$display(b, " ", i);
```

A) -1 -1

B) 255 -1

C) -1 255

D) Compilation error

Q5. What is printed?

```
int assoc[string];
assoc["key1"] = 10;
if (assoc.exists("key2")) $display("Found");
else $display("Not Found");
```

A) Found

B) Not Found

C) Compilation error

D) Runtime error

Q6. What does res contain?

```
int arr[] = '{1, 2, 3, 4, 5};
int res[] = arr.find(x) with (x > 3);
```

A) {4, 5}

B) {3, 4, 5}

C) {1, 2, 3}

D) Compilation error

Q7. What happens when this code runs?

```
class MyClass;
  int value = 10;
endclass

MyClass obj;
obj.value = 20;
```

A) obj.value is set to 20

B) Null pointer dereference error

C) Compilation error

D) obj is automatically created with new()

Q8. What is printed?

```
class Base;
  virtual function void show();
    $display("Base");
  endfunction
```

```
endclass

class Derived extends Base;
  function void show();
    $display("Derived");
  endfunction
endclass

Base b = new Derived();
b.show();
```

A) Base

B) Derived

C) Compilation error

D) Base Derived

Q9. What is output?

```
class Counter;
  static int count = 0;
  function new();
    count++;
  endfunction
endclass

Counter c1 = new();
Counter c2 = new();
$display(Counter::count);
```

A) 0

B) 1

C) 2

D) Compilation error

Q10. What is printed?

```
class Base;
endclass

class Derived extends Base;
  int x = 10;
endclass

Base b = new Derived();
Derived d;
if ($cast(d, b)) $display(d.x);
else $display("Cast failed");
```

A) 10

B) Cast failed

C) Compilation error

D) 0

Q11. What does solve a before b mean?

```
class Packet;
  rand int a, b;
  constraint c { solve a before b; a < b; }
endclass
```

A) a is randomized before b, allowing b's constraints to depend on a's value

B) a must always be less than b

C) b is randomized first

D) Both a and b are randomized simultaneously

Q12. What is the probability of level == 1?

```
class Packet;
  rand int level;
  constraint c { level dist {1 := 70, 2 := 20, 3 := 10}; }
endclass
```

A) 70%

B) 7%

C) 70 out of 100

D) Undefined (implementation-dependent)

Q13. What is the order of output?

```
initial begin
  fork
    begin
      #5 $display("A");
      #3 $display("B");
      #8 $display("C");
    end
    begin
      #3 $display("E");
      #5 $display("F");
      #6$display("G");
    end
  join_any
  #1 $display("Done");
end
```

A) A B C E F G Done

B) A E B F G Done

C) E A B F G Done C

D) E A B F G C Done

Q14. What is printed?

```
for (int i = 0; i < 5; i++) begin
  if (i == 2) continue;
  if (i == 4) break;
  $display(i);
end
```

A) 0 1 2 3 4

B) 0 1 3

C) 0 1

D) 1 2 3

Q15. What is printed?

```
task modify_val_1(int val);
  val = 100;
endtask

task modify_val_2(ref int val);
  val = 100;
endtask

int x = 5;
modify_val_1(x);
$display(x);
modify_val_2(x);
$display(x);
```

A) 5, 100

B) 100, 100

C) 100, 5

D) 105